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SUPERSEDING
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MILITARY SPECIFICATION

CALKING COMPOUND (LIQUID POLYMER POLYSULFIDE SYNTHETIC RUBBER, FORMULA No. 112, FOR METAL ENCLOSURES)

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments.

1. SCOPE

1.1 This specification covers calking compound for use in seams, joints, and dams formed between metal surfaces, to provide watertight and airtight seals. The compound is not intended for use in calking seams of wooden decks or hulls of vessels.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

RR-C-96 — Cans, Steel (Tinned-Plate, Terne-Plate and Black-Sheet); Friction-Covers.
TT-D-301 — Dibutyl-Phthalate; Plasticizer (for Use in Organic Coatings).
TT-P-143 — Paint, Varnish, Lacquer, and Related Materials; General Specification for Packaging, Packing, and Marking.
ZZ-R-601 — Rubber-Goods; General Specifications (Methods of Physical Tests and Chemical Analyses).
LLL-B-631 — Boxes; Fiber, Corrugated (for Domestic Shipment).
LLL-B-636 — Boxes; Fiber, Solid (for Domestic Shipment).

MILITARY

JAN-P-124 — Packaging and Packing for Overseas Shipment — Containers (Cans, Pails, and Drums), Metal (for Other Than Subsistence Items).
JAN-C-490 —Cleaning and Preparation of Ferrous Metal Surfaces for Organic Protective Coatings.
MIL-P-15328 —Pretreatment, Wash Primer (for Metals) (Formula No. 117).

NAVY DEPARTMENT

General Specifications for Inspection of Material.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 —Marking of Shipments.

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

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2.2 Other publications. — The following publication, of the issue in effect on date of invitation for bids, unless otherwise stated, forms a part of this specification:

CONSOLIDATED CLASSIFICATION COMMITTEE
Consolidated Freight Classification.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Formula.

3.1.1 The calking compound shall consist of ingredients conforming to the applicable specifications or equivalent to the commercial grades designated, and mixed in the proportions shown in table I.

TABLE I.—*Ingredients.*

Ingredients	Specification or commercial designation	Parts by weight
Component A		
Polysulfide, liquid polymer	Thiokol, LP–2, or equivalent	100
Carbon black, medium thermal	Thermax, or equivalent	100
(Ethyl) or (Methyl) phthalyl ethyl glycollate	Santicizer E–15 or Santicizer M17, or equivalent	50
Stearic acid	Triple press, rubber compounding quality	1
Component B (accelerator)		
Lead peroxide	Commercial grade 90 percent[1]	75
Dibutyl phthalate	Spec. TT–D–301	60
Stearic acid	Triple press rubber compounding quality	3

[1] Lead peroxide, commercial grade, should be equivalent in composition to the following analysis:

Lead Peroxide	90.5 percent.
Iron	0.035 percent.
Chlorides	60 parts per million.
Sulfates	120 parts per million.
Moisture	0.25 percent maximum.

3.2 Manufacture. — The component raw materials of the calking compound shall be mixed and ground as required to produce a product which is uniform, stable, free from grit, and entirely suitable for the purpose intended. The product shall be capable of being broken up with a paddle to a smooth, uniform consistency and shall not liver, thicken, curdle, gel, nor show any other objectionable properties.

3.2.1 *Preparation of component A.*—The carbon black shall be ground thoroughly into the other ingredients on a clean mill. Three passes on a three-roll mill normally are reare required. Any contamination with component B or other material shall be avoided.

3.2.2 *Preparation of component B.*—The lead peroxide shall be ground thoroughly into the other ingredients on a clean mill. Three passes on a three-roll mill noromally are required. Any contamination with component A or other material shall be avoided.

3.2.3 *Proportions.*—At the time of mixing, eight parts by weight of component B shall be combined with 92 parts by weight of component A.

3.3 Quantitative requirements.—The calking compound shall conform to the following:

3.3.1 *Uncured compound.*

3.3.1.1 *Working life.*—When the uncured compound is tested as specified in 4.4.1.1 the viscosity readings (using MacMichael viscosimeter) shall be within the following limits:

	Degrees
Initial	12 to 15
After 6 hours	16 to 24

3.3.1.2 *Curing time.* — When tested as specified in 4.4.1.2, the compounds shall become cured in no more than 3 days, but no less than 1 day under conditions of 75 percent relative humidity, and shall become cured in no more than 4 days, but no less

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than 2 days under conditions of 45 percent relative humidity.

3.3.2 *Cured compound.*—The properties of the cured material shall be as follows:

3.3.2.1 *Hardness.* — Shore A durometer shall be 25 ± 5 when tested in accordance with 4.4.2.1.

3.3.2.2 *Specific gravity.* — The specific gravity shall be 1.43 ± 0.03 when tested in accordance with 4.4.2.2.

3.3.3 *Stability of accelerator.* — When tested as specified in 4.4.3, the consistency of the accelerator shall remain unchanged and there shall be no gelling or solidifying after 96 hours of testing.

3.4 The 1-gallon pail of component A and the ½-pint can of component B specified in 5.1 shall be legibly marked by stencil, lithograph or securely affixed label with "MIL–C–15705, component A," or "component B" as applicable, and the following:

> "*Instruction for mixing.*—Contents of the container of component B shall be stirred to a smooth paste before mixing. One ½-pint can of component B shall be added to each 1-gallon can of component A and stirred until the mixture is uniform in color and no brown specks are visible. Mixing may be done by power stirrer, providing the temperature of the mixture does not reach 90° F., or higher. Temperatures above 80° F. greatly shorten the working life of the compound. Temperatures below 50° F. so lengthen the time before hardening that application of external heat may be necessary. This material shall be used as soon after mixing as possible. At normal temperatures, hardening usually takes place within 1 day after mixing, and therefore the material should not be left standing overnight before use."

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection procedures. — For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

4.2 Sampling.

4.2.1 *Lot.*—For purposes of sampling, a lot shall consist of component A from one batch and component B from one batch offered for delivery at one time.

4.2.2 *Sampling for lot acceptance tests.*

4.2.2.1 *Component A.*—From each lot, two separate 1-gallon samples from separate portions of the lot shall be taken by the Government inspector and submitted for tests specified in 4.4. The laboratory may perform complete tests on only one of the samples provided sufficient tests are made on the second to verify that both are alike.

4.2.2.2 *Component B.*—From each lot, two separate ½-pint samples from separate portions of the lot shall be taken by the Government inspector and submitted for tests specified in 4.4. The laboratory may perform complete tests on only one of the samples provided sufficient tests are made on the second to verify that both are alike.

4.2.2.3 *Testing laboratories.*—The samples taken by the Government inspector shall be tested at the place of manufacture under the supervision of the Government inspector, if adequate facilities are available. Otherwise, the samples shall be tested at a Government laboratory designated by the bureau or agency concerned.

4.2.3 *Sampling for inspection of containers.*—A random sample of filled containers shall be selected by the Government inspector in accordance with Standard MIL–STD–105 at inspection level I and acceptable quality level = 2.5 percent defective to verify compliance with all requirements regarding fill, closure, marking, and other requirements not involving tests.

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4.3 Inspection.

4.3.1 *Inspection of containers.* — Each sample filled container shall be examined for defects of construction of the container and the closure, for evidence of leakage, and unsatisfactory markings; each filled container shall also be weighed to determine the amount of contents. Any container in the sample having one or more defects or under required fill, shall be rejected, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of Standard MIL–STD–105 the lot represented by the sample shall be rejected. Rejected lots may be resubmitted for Government inspection provided the contractor has removed or repaired all nonconforming containers.

4.4 Tests.

4.4.1 *Uncured compound.* — A standard test batch shall be mixed, consisting of components A and B combined in the proportions specified in 3.2.3. The mixing shall be performed at room temperature (70 ± 5° F.) and shall be continued until the mixture is uniform and no brown specks are apparent. Immediately after mixing, the test batch shall be subjected to the following tests:

4.4.1.1 *Working life.*—Measurements of viscosity shall be conducted at room temperature (70 ± 5° F.), using a MacMichael viscosimeter operated at a speed of 20 revolutions per minute (r.p.m.) and using a number 20 wire immersed to a depth of 4 centimeters. A reading shall be taken immediately after mixing of the compound, and the result shall be recorded as the initial viscosity. The compound shall then be allowed to set for 6 hours at room temperature (70 ± 5° F.), after which another measurement shall be performed in the same manner as for the initial reading. This result shall be recorded as the viscosity after 6 hours setting time.

4.4.1.2 *Curing time.*—Samples of the test batch shall be poured into six rubber molds with cylindrical cavities, ½ inch high by 1⅝ inches inside diameter. Three of the molds shall be placed on a corrosion resisting steel supporting screen in a closed desiccator containing a saturated aqueous solution of potassium carbonate. Three molds similarly shall be placed in a desiccator containing a saturated aqueous solution of sodium chloride. At 73° F., the relative humidities in the desiccators containing potassium carbonate and sodium chloride will be 45 ± 2 percent and 75 ± 2 percent, respectively. One specimen from each desiccator shall be removed at 24-hour intervals and examined. If the compound is still tacky, it shall be considered not fully cured and the specimen shall be returned to the desiccator. When the specimen it found to be no longer tacky, it shall be considered cured. Determinations of curing time shall thus be made in triplicate, and the average of the three observed periods shall be reported as the curing time.

4.4.2 *Cured compound.* — Immediately upon completion of the tests specified in 4.4.1.2, the samples remaining from these tests shall be subjected to the following:

4.4.2.1 *Hardness.* — A type "A" Shore durometer shall be used to determine hardness of the samples cured in accordance with 4.4.1.2. Readings shall be instantaneous values.

4.4.2.2 *Specific gravity.*—Specific gravity shall be determined by the method specified in Specification ZZ–R–601.

4.4.3 *Stability of accelerator.*—A 20-gram sample of component B shall be placed in a test tube which shall be immersed in a constant temperature oil bath maintained at 158 ± 2° F. After 96 hours at this temperature, a visual determination of the consistency of the sample of accelerator shall be made.

4.5 Acceptance of lots.—Lots shall be accepted by the Government inspector only upon satisfactory completion of the tests

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conducted at the place of manufacture, or receipt of a satisfactory test report from the Government laboratory conducting the tests.

5. PREPARATION FOR DELIVERY

5.1 Packaging for domestic and overseas shipment.

5.1.1 Component A shall be packaged in 1-gallon round multiple friction top metal containers in accordance with Specification TT–P–143, except as specified in 5.1.3. Each 1-gallon can shall contain 10 ± 0.20 pounds of component A.

5.1.2 Component B shall be packaged in ½-pint round multiple friction top metal containers in accordance with Specification RR–C–96, except as specified in 5.1.3. Each ½-pint can shall contain 1 ± 0.03 pound of component B.

5.1.3 When tin or terneplate cans are used the enamel coating on the exterior of the cans will not be required for domestic shipment. When cans are furnished of blackplate they shall be either phosphate treated in accordance with grade 1 of Specification JAN–C–490, and then coated as specified in Specification JAN–P–124, or be given one coat of wash primer pretreatment complying with Specification MIL–P–15328 on the exterior and then coated on the exterior and interior as specified in Specification JAN–P–124.

5.1.4 Components A and B shall be further packaged as a single unit under one stock number in a fiberboard box in accordance with Specification LLL–B–631 or LLL–B–636. Component B shall be placed on top of the 1-gallon can and protected from movement by means of cell spacers or buffer pads formed from scored sheets of fiberboard. No material shall be accepted for shipment if more than 3 months old.

5.2 Packing for domestic and overseas shipment.—The units in fiberboard boxes packaged as specified in 5.1.4 shall be packed in accordance with Specification TT–P–143 for 1-gallon cans.

5.3 Marking.

5.3.1 Exterior containers shall be marked on and near the top in at least three locations with the following:

Name of material.

Stock number.

Specification number.

Name of manufacturer.

Manufacturer's batch number.

Date of manufacture.

Government contract or order number.

5.3.2 In addition to the marking specified in 3.4 and 5.3.1, and any additional marking specified in the contract or order, marking shall conform to Standard MIL–STD–129.

5.4 In addition to the requirements specified in 5.1, 5.2, and 5.3, the packaging, packing, and marking shall conform to the rules of the "Consolidated Freight Classification."

6. NOTES

6.1 Ordering data. — Procurement documents should specify the title, number, and date of this specification.

6.2 The formula shown in table I is designated Navy Standard Formula No. 112. Wherever formula No. 112 is specified, the compound shall conform to this specification.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way

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